

A Knowledge-Based System for Educational Assessment: Knowledge Engineering, Multi-Method Explainable AI, and Fairness-Aware Decision Support on PISA 2022 Data

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Abstract

Background: Knowledge-based systems that integrate domain knowledge with machine learning remain rare in large-scale educational assessment, despite the field’s need for reproducible, interpretable, and fairness-aware analytical pipelines.

Methods: We develop a knowledge-based system that organizes 33 PISA 2022 predictors (613,744 students, 80 countries) across individual, interaction, institutional, and contextual levels, and classifies ICT variables by access, skills, and usage dimensions. Tree-based ensembles are trained with Bayesian hyperparameter optimization (Optuna, 50 trials), interpreted through SHAP, permutation importance, and Accumulated Local Effects (with LIME examined as a supplementary comparison), and evaluated against Equalized Odds, Demographic Parity, and ABROCA. The full pipeline is reproducible on a single workstation.

Results: Tuned XGBoost achieves weighted RMSE = 54.10 ($R^2 = 0.681$)

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and $AUC = 0.903$ (Brier = 0.126), a 23% gain over linear baselines. Supplementary k -fold CV confirms negligible partition sensitivity ($AUC = 0.890 \pm 0.001$). ICT resources and ICT self-efficacy consistently rank among the strongest predictors. SES disparities are the largest fairness concern (ABROCA = 0.023), with pronounced disadvantage for low-SES immigrant-background students ($AUC = 0.831$ vs. 0.903). Country-group holdout ($AUC = 0.847$) confirms limited cross-context generalization.

Conclusion: The proposed system offers a reproducible, fairness-aware template for knowledge-driven educational assessment; its modular architecture supports future ablation studies of the knowledge engineering layer without pipeline redesign.

Keywords: explainable artificial intelligence, knowledge-based system, educational assessment, algorithmic fairness, machine learning, PISA

1. Introduction

Knowledge-based systems that integrate domain knowledge with machine learning are increasingly needed to support human prediction and decision-making in complex, data-rich environments. Three knowledge engineering obligations arise in developing such systems: representing domain knowledge in a form the machine learning workflow can exploit, extracting and validating knowledge from fitted models, and encoding theoretical constructs into operational feature structures. Educational assessment at global scale—data spanning 80+ countries with heterogeneous systems, infrastructures, and socioeconomic conditions—imposes these obligations with particular force. The predictors of student achievement are not merely statistical variables

but theoretically grounded constructs whose relationships embody domain knowledge from educational psychology and digital inequality research.

Mathematics literacy is a core indicator of students’ quantitative reasoning capacity. Understanding which factors predict mathematics achievement—and how digital learning technologies contribute—is a central concern for educational technology research [1]. PISA 2022, conducted in the aftermath of COVID-19-related school disruptions, places mathematics as the major assessment domain and provides detailed ICT-related questionnaire variables, offering a particularly informative setting [2, 3].

The relationship between digital technology and educational outcomes resists simple characterization. Recent research reveals an “ICT paradox”: digital access and literacy positively predict achievement, whereas certain forms of advanced digital skills and unstructured technology use show negative associations [4]. This complexity calls for structured approaches that distinguish among access, skills, usage purposes, and learning outcomes rather than treating technology as a monolithic construct [5, 6].

In parallel, educational data mining increasingly applies machine learning to predict achievement [7]. Tree-based ensembles capture nonlinear predictive structure that linear models may miss [8], while SHAP explanations reveal which variables fitted models rely upon [9]. However, developing a knowledge-based system for large-scale assessment data requires satisfying obligations beyond model comparison: proper knowledge representation conventions governing the use of plausible values, survey weights, and replicate weights [10, 11]; a knowledge extraction and validation framework that interprets model explanations as fitted-model patterns rather than causal evidence [12,

13]; and domain knowledge encoding that organizes digital-learning variables by theoretical structure rather than treating them as an undifferentiated list.

We address these challenges by developing a knowledge-based ML system that integrates domain knowledge from educational research and digital inequality scholarship into a reproducible workflow. Drawing on ecological systems theory [14] and digital divide research [6], the system organizes 33 predictors across multiple ecological levels and classifies ICT variables by access, skills, and usage dimensions. Tree-based ensembles are trained with Bayesian hyperparameter optimization, interpreted through a multi-method knowledge extraction framework (SHAP, permutation importance, and Accumulated Local Effects, with LIME as a supplementary comparison), and evaluated against formal algorithmic fairness metrics—all from a single workstation with fixed random seeds and deterministic explanation samples.

The present study addresses four research questions:

1. Which features are most predictive of PISA 2022 mathematics literacy when organized under a knowledge-based hierarchical feature framework?
2. Do tree-based ensemble models with Bayesian hyperparameter optimization outperform traditional linear baselines by a clear margin on global-scale educational data?
3. Does the multi-method knowledge extraction framework produce convergent evidence across explanation methods (SHAP, permutation importance, LIME, and Accumulated Local Effects), and do the resulting feature importance patterns align with domain knowledge?
4. Are predictive performance and model explanations stable across demo-

graphic subgroups, socioeconomic strata, and country contexts?

This work is positioned as a knowledge-based systems contribution rather than a conventional educational data mining study. The system encodes domain expertise from ecological systems theory and digital divide research into structured feature engineering; employs tree-based ensembles tuned via Bayesian optimization as its predictive core; integrates four complementary XAI methods with cross-method validation to ensure explanation reliability; and delivers fairness metrics and robustness benchmarks actionable for educational stakeholders. The full pipeline operates from a single workstation with fixed random seeds and deterministic explanation samples, satisfying the reproducibility standard for knowledge-driven analytical systems [8, 7]. Section 4.7 provides the full system architecture description. The term "knowledge extraction" in this paper refers to post-hoc explanation synthesis from fitted models, distinct from symbolic rule extraction but methodologically aligned with KBS's interest in making model-internal patterns externally interpretable [15].

2. Knowledge Engineering Framework

The framework structures raw PISA variables into theoretically grounded feature groups through a systematic knowledge encoding process. Following the knowledge engineering paradigm—knowledge acquisition, knowledge representation, and knowledge validation—we first acquire domain knowledge from ecological systems theory and digital divide research, then represent it as a hierarchical feature organization supplemented by an ICT taxonomy, and finally validate the encoded knowledge through the testable predictions

of Section 2.3. The framework functions as an organizational tool rather than a formal causal model, and the ML models remain nonparametric and predictive.

2.1. Framework description

Drawing on ecological systems theory [14, 16], we organize 33 predictors across four nested hierarchical levels:

- **Individual level** — student characteristics, family resources (home possessions, parental occupation), and daily interactions with teachers and peers; proximal factors shaping learning opportunities and motivational beliefs such as self-efficacy [17].
- **Interaction level** — connections between settings: parent-school communication, alignment between home learning resources and school curriculum, and the association between family SES and school quality.
- **Institutional level** — school-level factors (school climate, resource shortages, teacher and student behavioral climate) and community ICT infrastructure.
- **Contextual level** — national education policies, digital infrastructure, and socioeconomic inequality; country identity serves as a proxy.

Grounded in digital inequality research [5, 6], ICT-related PISA variables are classified into a three-level taxonomy: **access** (ICT resources at home, ICTRES; ICT availability at school, ICTSCH; home ICT availability, ICTHOME), **skills/confidence** (ICT self-efficacy, ICTEFFIC), and **usage** (subject-related

ICT use, ICTSUBJ; ICT information behavior, ICTINFO; study-related ICT use, STUDYHMW).

This dual organization informs three design decisions: (1) feature coverage spanning all four hierarchical levels; (2) disaggregated ICT analysis by taxonomy level; and (3) context-sensitivity evaluation via country fixed-effects and country-group holdout checks. Figure 1 provides a visual synthesis.

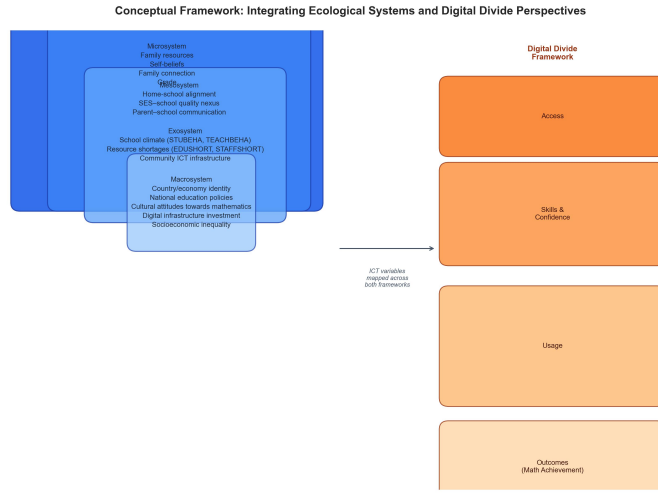


Figure 1: Knowledge engineering framework: hierarchical feature organization (individual, interaction, institutional, and contextual levels) integrated with the ICT feature taxonomy (access, skills/confidence, and usage dimensions) for 33 PISA 2022 predictors across 80 countries

2.2. Testable predictions

The framework yields six testable predictions, evaluated in Section 6 (Table 9):

P1 — Proximal dominance: Individual-level features are the strongest predictors.

- P2 — Institutional contribution:** School-level features contribute predictive information beyond what individual and family characteristics alone provide.
- P3 — Contextual moderation:** Country identity moderates lower-level effects; models incorporating country information outperform those that do not.
- P4 — Skill-access gradient:** ICT self-efficacy carries greater predictive weight than ICT resource availability.
- P5 — Usage quality gradient:** Subject-related ICT use carries greater predictive weight than general ICT information behavior.
- P6 — Cross-context heterogeneity:** Feature importance rankings vary across country contexts.

These are statements about expected patterns in fitted-model behavior, not causal hypotheses.

2.3. Knowledge encoding protocol

The framework follows three knowledge engineering stages:

Knowledge acquisition. Domain knowledge is acquired from two established theoretical sources: ecological systems theory [14, 16], which situates student achievement within nested environmental layers (individual, interaction, institutional, contextual), and digital divide research [5, 6], which distinguishes ICT access, skills, and usage as qualitatively distinct dimensions. These sources are operationalized to the PISA 2022 variable set through a structured mapping procedure documented in Supplementary Table S6.

Knowledge representation. The acquired knowledge is represented as a two-part structure: a four-level hierarchical feature organization (Section 2.1) and a three-level ICT taxonomy (Section 2.1). Each of the 33 predictors is assigned to one ecological level and—if ICT-related—to one taxonomy dimension. The representation is declarative (features carry theoretically grounded type labels) rather than procedural (no causal graph or inference rules are specified), consistent with the system’s predictive rather than causal scope.

Knowledge validation. The encoded knowledge is validated through six testable predictions (P1–P6, Section 2.3) that are evaluated against fitted-model behavior in Section 6. This structured knowledge validation protocol makes domain assumptions explicit and empirically falsifiable, a defining characteristic of knowledge-based systems [18, 19].

3. Related Work

3.1. Knowledge extraction and explanation in large-scale assessment

Gradient boosting models including XGBoost and LightGBM have become standard tools in educational data mining, capturing nonlinearities and interactions that linear baselines miss [20, 21, 22]. Recent reviews identify insufficient attention to complex sampling designs and missing-data mechanisms [23] and underexplored explainability and fairness [24]. A 2026 systematic review of 72 EDM studies confirmed SHAP as the dominant XAI method and identified interpretability-fairness integration as a critical underexplored area [25]; a parallel systematic literature mapping corroborated that interpretability is rarely treated as a core design objective [26]. Recent PISA-ML

studies have applied these methods with SHAP interpretation to East Asian systems [27], cross-national resilience analysis [28], Spanish students [29], U.S. data [30], and Latin American low-performer identification [31, 32]; Öz and Bulut [33] demonstrated ensemble advantages on the full 80-country sample but included neither XAI nor fairness evaluation. A metaheuristic-optimized LightGBM system with SHAP analytics for student performance prediction [34] and an EBM-based dropout prediction study documenting global and local explainability trade-offs [35] confirm the trend toward interpretable ML architectures (Supplementary Table S3 provides a structured comparison).

SHAP and permutation importance are widely used to summarize predictor contributions [9, 13], and post-hoc explanation methods are increasingly preferred in educational applications [36]. Recent work extends SHAP-based explanation to model uncertainty quantification, demonstrating the method’s flexibility across domains [37]. Within the Knowledge-Based Systems literature, Caro-Martinez et al. [15] proposed a case-based reasoning platform for designing explanation experiences, demonstrating how knowledge-based approaches can systematize XAI method selection—a concern shared by this study. Because different interpretability algorithms produce complementary but divergent rankings [38] and correlated predictors can distort marginal feature effect estimates [39], multi-method comparison is advisable.

3.2. Algorithmic fairness in predictive modeling

Predictive models can perpetuate inequities when fairness is not explicitly evaluated [40, 41], and recent reviews recommend expanding fairness evaluations beyond gender and race to socioeconomic and intersectional subgroups [42]. Mirsch et al. [43] demonstrated that intersectional fairness analysis

reveals disparities invisible to single-axis evaluation, reinforcing the need for multi-dimensional subgroup assessment. Within recent KBS publications, Zhang et al. [44] proposed a causal path approach to fairness-aware feature selection, Wang and Luo [45] introduced an object-relational causal inference method for detecting latent discrimination, and Pei [46] presented adaptive regularization for fair federated learning, reflecting the journal’s growing emphasis on algorithmic fairness. In the educational domain, Pham et al. [18] recently proposed fairness-enhancing methods for educational ML, while Gándara et al. [47] found common bias-mitigation techniques often ineffective, pointing to transparent subgroup evaluation as a minimum requirement.

3.3. Research gap and present contribution

Five gaps remain, each addressed by this study: (1) most PISA-ML studies rely on single-country or regional samples—we use the full 80-country PISA 2022 sample with proper survey-weight and plausible-value conventions; (2) no prior study systematically compares multiple XAI methods within a single analytical framework—we compare SHAP, permutation importance, LIME, and ALE with rank correlation validation; (3) formal algorithmic fairness metrics are largely absent from PISA-ML studies—we implement Equalized Odds, Demographic Parity, and ABROCA [48] with intersectional subgroup analysis; (4) ICT variables are typically treated as supplementary covariates—we make them central analytical targets within the knowledge engineering framework and ICT taxonomy of Section 2; and (5) no prior study has structurally validated domain-knowledge-derived predictions against fitted-model behavior—we evaluate six such predictions in Section 6.

4. Knowledge-Based System Architecture and Methodology

4.1. Data and preprocessing

The system ingested public-use PISA 2022 student and school questionnaire data from the OECD PISA 2022 Database [3]. The processed analysis frame comprises 613,744 students from 80 countries and economies. Student and school questionnaires were merged by country and school identifier. Four school-level variables entered the model: student behavior hindering learning (STUBEHA), teacher behavior hindering learning (TEACHBEHA), educational material shortage (EDUSHORT), and staff shortage (STAFFSHORT). A complete code-to-description mapping is provided in Supplementary Table S6. Figure 2 summarizes the full analytical pipeline.

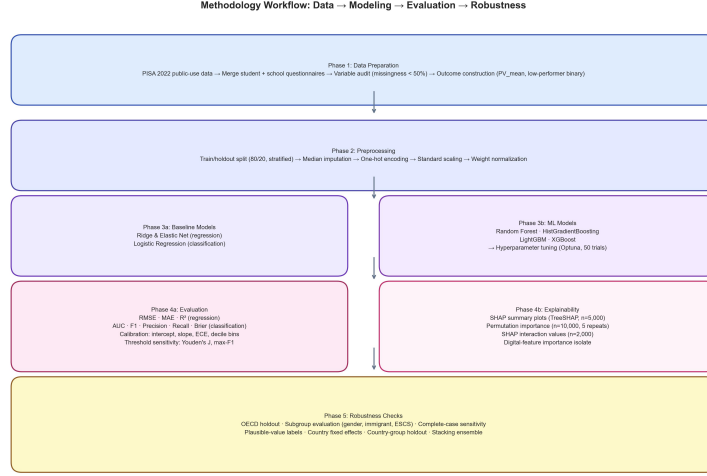


Figure 2: Knowledge-based system architecture: from PISA 2022 public-use data through the five-layer pipeline—(1) knowledge acquisition (data ingestion and variable audit), (2) knowledge representation (hierarchical feature organization and ICT taxonomy), (3) inference engine (Bayesian HPO and ensemble training), (4) knowledge extraction (multi-method XAI with cross-method validation), and (5) decision support and fairness audit (fairness metrics, intersectional analysis, and eight robustness checks). All components operate on a single workstation with fixed random seeds.

The weighted global mathematics mean was 424.29 ($SE = 0.72$ using Balanced Repeated Replication [BRR] replicate weights and plausible-value pooling). The weighted low-performer rate based on plausible-value pooling was 53.30% ($SE = 0.35$), while the modeling label based on the row-wise plausible-value mean produced a weighted rate of 53.90% ($SE = 0.35$). These estimates distinguish population descriptive quantities from modeling outcomes.

4.2. Outcomes and feature set

The continuous outcome is mathematics literacy, operationalized as the row-wise mean of PV1MATH through PV10MATH. The binary outcome is low-performing student status, defined as mathematics performance below 420.07—the lower bound of PISA proficiency Level 2 [49].

A reproducible variable audit was conducted prior to modeling. Of 36 configured predictors available in the processed data, 33 were retained with missingness at or below 50%. Two parent-school relationship variables, PQSCHOOL and PASCHPOL, were excluded because both exceeded 85% missingness. Missing values were handled by median imputation for numeric variables and most-frequent imputation for categorical variables. Categorical variables were one-hot encoded with rare-category grouping. The same frozen feature set was used for both regression and classification tasks.

4.3. Model training and hyperparameter optimization

Traditional baselines comprised ridge regression, elastic net, and L2-regularized logistic regression. ML models included random forest, histogram gradient boosting, XGBoost, and LightGBM, all with a fixed random seed (20260510). Deep learning baselines were excluded because tree-based ensembles consistently outperform neural networks on tabular data with heterogeneous feature types [50, 23]. Intrinsically interpretable models (EBMs, GAMs) were not included as primary baselines; a preliminary benchmark confirming their competitiveness is reported in Section 7.

Hyperparameters for XGBoost and LightGBM were tuned using Optuna (Bayesian optimization, 50 trials per model-task), minimizing RMSE (regression) and log-loss (classification) on a 15% validation subset. Fifty trials was

chosen based on preliminary convergence diagnostics showing diminishing returns beyond 40 trials; search space coverage was verified via optimization history logging. The full search space is documented in Supplementary Table S7.

A stacking ensemble combined random forest, LightGBM, and XGBoost as base learners with ridge or logistic regression as the meta-learner, using out-of-fold predictions from 5-fold cross-validation [29].

The primary evaluation used a stratified 80/20 holdout. Bootstrap 95% CIs from 1,000 resamples of holdout predictions confirmed narrow intervals (RMSE 54.10 [53.58, 54.66], AUC 0.903 [0.898, 0.907]). A supplementary 5-fold CV with default-parameter models yielded near-zero cross-fold variance (XGBoost AUC = 0.890 ± 0.001), confirming negligible partition sensitivity at this sample size. The full pipeline ran on a workstation (Apple M5, 12 cores, 16 GB unified memory) in approximately 12 hours.

4.4. Multi-method XAI framework

SHAP. TreeSHAP on a deterministic 5,000-row explanation sample; cross-model consistency (XGBoost vs. LightGBM, Spearman’s $\rho = 0.83$) and near-perfect explanation fidelity ($R^2 = 0.9999$ between predictions and SHAP value reconstruction) support stability at this sample size. SHAP interaction values were computed for theoretically motivated feature pairs (ICT resources $\times$ home possessions, ICT self-efficacy $\times$ mathematics self-efficacy).

Permutation importance. Deterministic 10,000-row sample with five repeats, using AUC (classification) and RMSE (regression) as scoring metrics.

ALE. First-order ALE on a 10,000-row sample with 20 bins, with second-order ALE for key feature pairs. ALE is preferred over Partial Dependence

when features are correlated [13, 39].

LIME. Local explanations on a 500-instance sample for the best classification model [51], included as a supplementary comparison given its known sensitivity to correlated features.

Cross-method validation. Feature importance rankings from all methods were compared using Spearman’s ρ and Kendall’s τ rank correlation, with explanation fidelity and faithfulness computed following Kışman et al. [52].

4.5. Fairness evaluation

Formal fairness metrics were computed for the best classification model across three demographic dimensions: gender, immigrant background, and socioeconomic status as measured by the PISA Economic, Social and Cultural Status (ESCS) index quartile groups. The metrics included Equalized Odds Difference, Demographic Parity Difference, and ABROCA (Absolute Between-ROC Area) [48]. Intersectional subgroup analysis further examined combined disparities, for instance low-SES students with non-native immigrant backgrounds.

4.6. Robustness evaluation

Eight robustness checks were implemented: (1) OECD-only holdout evaluation; (2) subgroup holdout by gender, immigrant background, and ESCS quintiles; (3) complete-case sensitivity analysis; (4) recomputation of low-performing status using each plausible value separately; (5) country fixed-effect comparison on a 120,000-row robustness sample; (6) country-group holdout (training on 64 countries, evaluation on 16); (7) explanation stability assessment across demographic subgroups using pairwise Spearman’s ρ ; (8)

temporal stability analysis comparing feature importance rankings from PISA 2022 against those from PISA 2018, using harmonized variable definitions and Spearman’s ρ as the pre-specified stability metric.

4.7. Knowledge-based system architecture overview

The system adopts a five-layer knowledge-based system architecture aligned with classical KBS components:

- **Layer 1: Knowledge acquisition layer** — validates and merges PISA public-use files, documenting variable provenance, missingness, and ecological-level assignment (Supplementary Table S6); corresponds to the knowledge acquisition stage in the KBS knowledge engineering cycle.
- **Layer 2: Knowledge representation layer** — applies the hierarchical feature organization and ICT taxonomy derived from ecological systems theory and digital divide research; corresponds to the knowledge representation stage, encoding domain expertise as structured feature type labels.
- **Layer 3: Inference engine** — Bayesian hyperparameter optimization (Optuna, 50 trials), tree-based ensemble training (XGBoost, LightGBM, random forest), stacking, and stratified holdout evaluation; tuned XGBoost serves as the primary inference engine.
- **Layer 4: Knowledge extraction and explanation layer** — SHAP, permutation importance, LIME, and ALE with cross-method validation

(Spearman’s ρ , Kendall’s τ); extracts and validates knowledge from fitted models.

- **Layer 5: Decision support and fairness audit layer** — formal fairness metrics (Equalized Odds, Demographic Parity, ABROCA), intersectional analysis, and eight robustness checks; produces actionable decision-support output for educational stakeholders.

The pipeline deploys on a single workstation (Apple M5, see Section 4.3). All components are accessible through a public benchmark repository with fixed random seeds and deterministic explanation samples.

The knowledge representation layer (Layer 2) encodes domain expertise through theoretically motivated feature selection and taxonomic classification. The hierarchical feature organization (Section 2.1) structures 33 predictors across four ecological levels; the ICT taxonomy (Section 2.1) disaggregates seven digital-learning variables by access, skills/confidence, and usage dimensions. Together with the structured proposition validation (P1–P6), these components form a knowledge-based architecture in which domain knowledge constrains and guides the ML workflow, from feature selection through interpretation, rather than treating the pipeline as a purely data-driven exercise.

4.8. Knowledge validation protocol

The six domain-knowledge-derived propositions (P1–P6, Section 2.3) constitute a knowledge validation protocol that makes domain assumptions explicit and empirically testable. Each proposition specifies: (1) the knowledge basis from which it is derived (ecological systems theory or ICT taxonomy);

(2) the empirical test that operationalizes it (e.g., SHAP rank comparison for P4, country-group holdout for P3); (3) the expected pattern if the encoded knowledge is consistent with fitted-model behavior; and (4) a graded assessment of support (strongly supported, supported, or not supported). The protocol is evaluated in Section 6 (Table 9).

5. Experimental Results

5.1. *Sample and variable audit*

The knowledge acquisition layer processed 613,744 students from 80 countries, organizing 33 predictors across four hierarchical levels (individual, interaction, institutional, contextual), with ICT variables classified by the three-level access–skills–usage taxonomy (Section 2). The processed analytic frame (Table 1) showed pronounced cross-national variation in weighted low-performer rates (below 25% to above 70%; Table 2), motivating the country fixed-effect and country-group holdout checks.

Table 1: Analysis frame summary: PISA 2022 student and school questionnaire data after merge and variable audit

Characteristic	Value
Students	613,744
Countries/economies	80
Weighted mathematics mean (plausible-value pooled)	424.29
Weighted low-performer rate (pooled)	53.9%
Unweighted low-performer rate	45.5%
Available configured features	36
Main model features (retained)	33
Extended features (incl. robustness)	34
School-questionnaire variables	4 (see text)

Note. The descriptive estimates in this table use the row-wise mean of mathematics plausible values as the modeling outcome. Population-level descriptive estimates (BRR and plausible-value pooled) are reported in Table 2. Only 6.47% of students ($n = 39,706$) had complete data on all 33 predictors (complete-case low-performer rate 37.88% vs. 45.54% in the full sample), confirming that complete-case analysis would substantially alter the analyzed population; median/mode imputation was therefore applied. School-level variables are STUBEHA, TEACHBEHA, EDUSHORT, and STAFFSHORT (see Supplementary Table S6 for code-to-description mapping).

Table 2: Weighted descriptive estimates using BRR replicate-weight and plausible-value standard errors

Estimate	Mean	SE	95% CI lower	95% CI upper
Math. score mean (PV-pooled)	424.29	0.72	422.88	425.70
Low-performer rate (PV-pooled)	53.30%	0.35 pp	52.62%	53.98%
Low-performer rate (model label)	53.90%	0.35 pp	53.22%	54.58%

Note. $n = 613,744$ students from 80 countries/economies. Estimates use final student weights (W_FSTUWT), 80 BRR replicate weights, and 10 mathematics plausible values (PV1MATH–PV10MATH). The model label is the row-wise mean of the 10 plausible values. PV-pooled estimates follow PISA multiple-imputation and replicate-weight conventions.

5.2. Model performance

The inference engine (tuned XGBoost) achieved weighted RMSE = 54.10 ($R^2 = 0.681$) for score prediction and AUC = 0.903 (Brier = 0.126) for low-performer classification, a 23% RMSE reduction over ridge regression (RMSE = 70.13). Tuned LightGBM performed comparably (RMSE = 54.15, AUC = 0.902); full results are reported in Tables 3 and 4. A three-learner stacking ensemble underperformed the best single model (RMSE = 57.07), consistent with prior findings that heterogeneous base learners can dilute meta-learner signal on tabular data.

Table 3: Weighted holdout regression performance for mathematics score prediction, main feature set

Model	RMSE	MAE	R ²
<i>Default-parameter models</i>			
Ridge regression	70.13	55.61	0.463
Elastic net	70.13	55.62	0.463
Random forest	62.72	49.38	0.571
Hist. gradient boosting	57.69	45.16	0.637
XGBoost	59.81	46.91	0.609
LightGBM	57.61	45.12	0.638
<i>Hyperparameter-optimised models</i>			
XGBoost (tuned)	54.10	42.37	0.681
LightGBM (tuned)	54.15	42.48	0.680
Stacking ensemble	57.07	44.72	0.644

Note. $n_{\text{train}} = 490,995$, $n_{\text{test}} = 122,749$, weighted by normalized W_FSTUWT. Models use 33 main features with median/mode imputation and one-hot encoding. Tuned hyperparameters obtained via Bayesian optimisation (Optuna, 50 trials per model-task) on a 15% validation subset; see Supplementary Table S7 for default vs. tuned values. Stacking ensemble: RF + LightGBM + XGBoost base learners, ridge meta-learner, 5-fold CV. Bold values denote best overall performance.

Table 4: Weighted holdout classification performance, main feature set, default threshold 0.50

Model	AUC	Avg. Prec.	F1	Precision	Recall	Brier
<i>Default-parameter models</i>						
Logistic (L2)	0.840	0.853	0.789	0.748	0.835	0.164
Random forest	0.870	0.882	0.808	0.777	0.841	0.148
Hist. gradient boosting	0.889	0.902	0.817	0.809	0.824	0.135
XGBoost	0.882	0.895	0.813	0.800	0.826	0.139
LightGBM	0.890	0.903	0.818	0.811	0.826	0.134
<i>Hyperparameter-optimised models</i>						
XGBoost (tuned)	0.903	0.916	0.828	0.822	0.834	0.126
LightGBM (tuned)	0.902	0.915	0.827	0.820	0.833	0.126
Stacking ensemble	0.894	0.907	0.821	0.814	0.828	0.131

Note. $n_{\text{train}} = 490,995$, $n_{\text{test}} = 122,749$, weighted by normalized W_FSTUWT. Low-performer threshold: mathematics score < 420.07 (below PISA Level 2). AUC = area under the ROC curve; Avg. Prec. = average precision; Brier = Brier score. Tuned hyperparameters obtained via Bayesian optimisation (Optuna, 50 trials per model-task); see Supplementary Table S7. Stacking ensemble: RF + LightGBM + XGBoost base learners, logistic meta-learner, 5-fold CV. Bold values denote best overall performance.

The best classification model was well-calibrated (calibration slope 0.987, ECE 0.008; Table 5) and robust across decision thresholds ($F1 \geq 0.74$ across three rules; Table 6).

Table 5: Calibration diagnostics for the best classification model (tuned XGBoost, AUC = 0.903)

Metric		Value				
Brier score		0.126				
Mean predicted probability		0.534				
Observed low-performer rate		0.539				
Calibration intercept		0.0469				
Calibration slope		0.987				
Expected calibration error (ECE)		0.0077				

Bin	Pred. range	n	Mean pred.	Observed	Gap
1	[0.0000, 0.1]	29,856	0.036	0.037	0.0001
2	[0.1, 0.2]	12,174	0.148	0.172	0.025
3	[0.2, 0.3]	9,256	0.249	0.256	0.006
4	[0.3, 0.4]	8,129	0.350	0.390	0.041
5	[0.4, 0.5]	7,955	0.450	0.462	0.012
6	[0.5, 0.6]	7,669	0.549	0.542	0.007
7	[0.6, 0.7]	7,839	0.651	0.653	0.002
8	[0.7, 0.8]	9,098	0.753	0.746	0.007
9	[0.8, 0.9]	11,340	0.854	0.855	0.001
10	[0.9, 1.0]	19,433	0.959	0.963	0.004

Note. Calibration metrics evaluated on the weighted classification holdout set ($n = 122,749$). ECE computed across 10 equal-width probability bins weighted by bin sample proportion. The calibration intercept and slope are from a logistic calibration model with predicted logits as the sole predictor.

Table 6: Classification threshold sensitivity across three decision rules

Model	Rule	Thresh.	F1	Precision	Recall	Brier
Logistic (L2)	Default (0.50)	0.50	0.789	0.748	0.835	0.164
	Youden’s J	0.59	0.774	0.787	0.760	0.164
	Max F1	0.46	0.791	0.732	0.861	0.164
Random forest	Default (0.50)	0.50	0.808	0.777	0.841	0.148
	Youden’s J	0.58	0.792	0.819	0.766	0.148
	Max F1	0.46	0.810	0.754	0.877	0.148
Hist. GBM	Default (0.50)	0.50	0.817	0.809	0.824	0.135
	Youden’s J	0.59	0.805	0.845	0.768	0.135
	Max F1	0.35	0.825	0.754	0.913	0.135
XGBoost	Default (0.50)	0.50	0.813	0.800	0.826	0.139
	Youden’s J	0.55	0.806	0.822	0.790	0.139
	Max F1	0.41	0.819	0.765	0.881	0.139
LightGBM	Default (0.50)	0.50	0.818	0.811	0.826	0.134
	Youden’s J	0.58	0.808	0.839	0.779	0.134
	Max F1	0.38	0.827	0.766	0.898	0.134

Note. Default threshold = 0.50. Youden’s J maximizes (sensitivity + specificity – 1). Max F1 maximizes the harmonic mean of precision and recall. All metrics are weighted by normalized W_FSTUWT on the holdout set ($n = 122,749$).

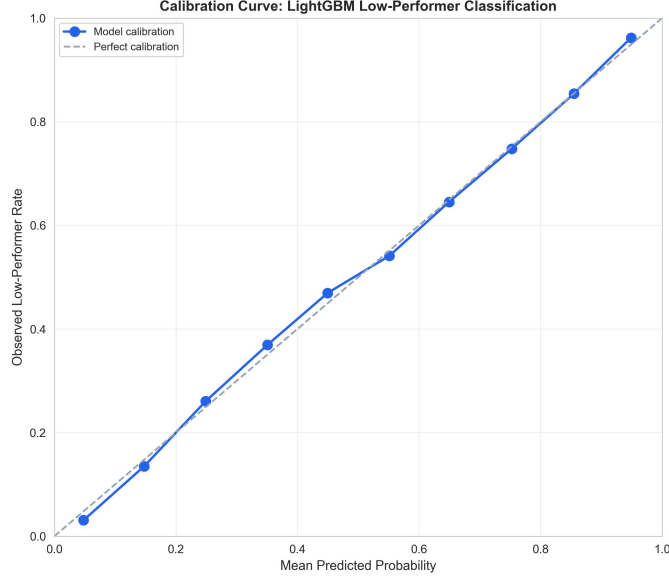


Figure 3: Calibration curves for the best classification model (tuned XGBoost, AUC = 0.903, Brier = 0.126). Predicted probability of low mathematics performance is plotted against observed low-performer rates across ten decile-style bins. The diagonal reference line indicates perfect calibration.

5.3. Global feature importance across XAI methods

The knowledge extraction layer’s SHAP and permutation importance analyses both identified home possessions (HOMEPOS) and mathematics self-efficacy (MATHEFF) as the two strongest predictors, with individual-level features dominating the hierarchy (Figures 4 and 5; SHAP cross-model consistency Spearman’s $\rho = 0.83$). The prominence of school-level variables (student and teacher behavior, resource shortages) confirms institutional-level predictive contribution beyond individual characteristics, as predicted by the hierarchical feature organization.

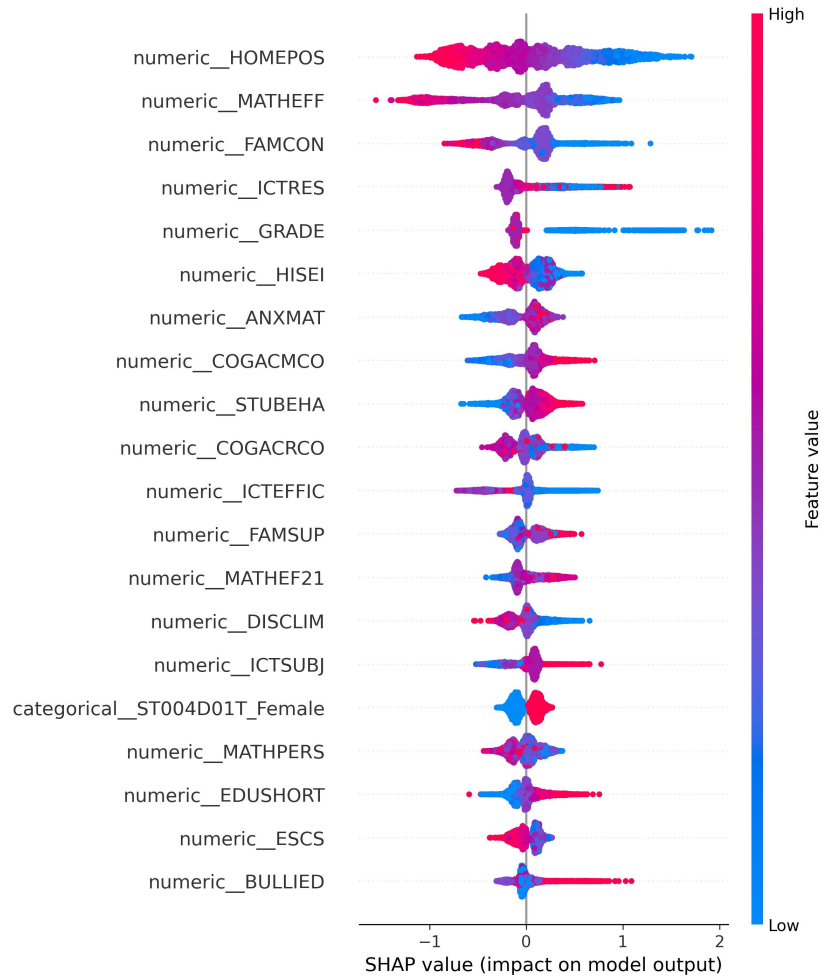


Figure 4: SHAP summary for the default-parameter LightGBM low-performer classification model ($AUC = 0.890$; the tuned LightGBM achieves $AUC = 0.902$), showing fitted-model feature contributions for 33 PISA 2022 predictors across a deterministic 5,000-student explanation sample. Red indicates high feature values; blue indicates low. Interpret as predictive model explanation, not causal evidence.

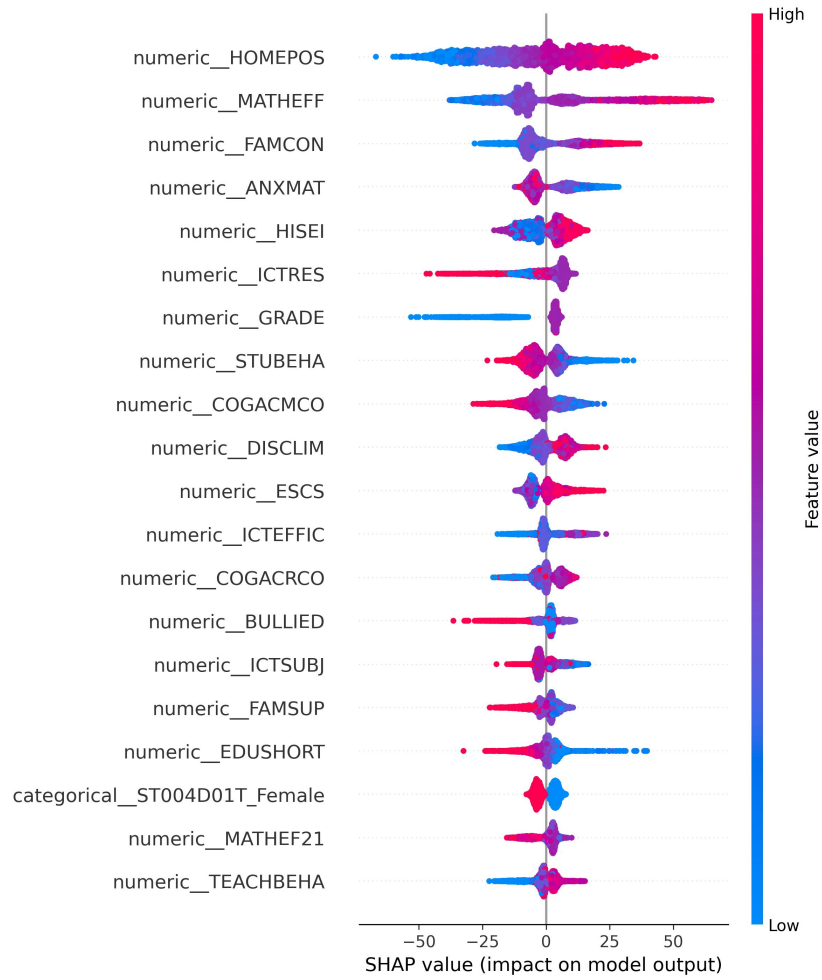


Figure 5: SHAP summary for the default-parameter LightGBM mathematics-score regression model ($\text{RMSE} = 57.61$, $R^2 = 0.638$; the tuned LightGBM achieves $\text{RMSE} = 54.15$), showing fitted-model feature contributions for 33 PISA 2022 predictors across a deterministic 5,000-student explanation sample. Red indicates high feature values; blue indicates low. Interpret as predictive model explanation, not causal evidence.

5.4. ICT feature analysis

The knowledge representation layer’s ICT taxonomy made visible that ICTRES and ICTEFFIC were the most important digital-learning predictors in both tasks; ICTSUBJ and STUDYHMW ranked next, while ICTINFO, ICTHOME, and ICTSCH had smaller importance values (Figure 6). This pattern—visible precisely because the ICT taxonomy disaggregates seven ICT variables by theoretical level—matches the taxonomy’s prediction that skills and confidence carry greater predictive weight than device access alone.

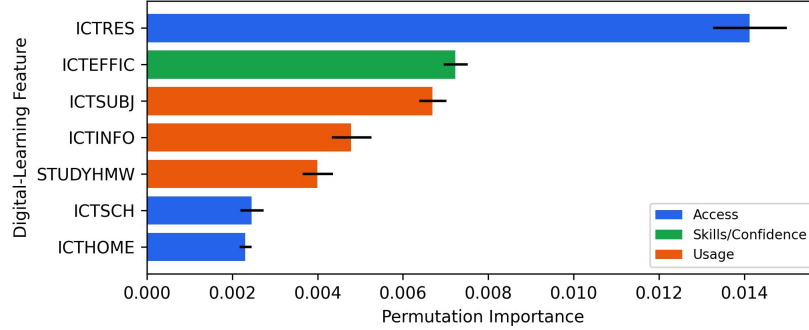


Figure 6: Permutation importance of ICT-related predictors from the tuned LightGBM classification and regression models, isolating the access, skills/confidence, and usage dimensions from the 33-predictor model. Tuned XGBoost (the system’s primary inference engine) achieves $AUC = 0.903$ and $RMSE = 54.10$; LightGBM achieves comparable performance with $AUC = 0.902$ and $RMSE = 54.15$.

5.5. Robustness and cross-context generalization

Subgroup holdout evaluation

Performance was broadly stable by gender ($AUC = 0.901$ female, 0.904 male) and declined for immigrant-background and low-SES students (Table 7).

Table 7: Subgroup holdout performance for the best classification model (tuned XGBoost) by gender, immigrant background, and ESCS quintile

Group	Value	n	AUC	F1	Prec.	Recall	Brier	RMSE
Gender	Female	60,761	0.901	0.832	0.831	0.833	0.127	51.73
	Male	61,972	0.904	0.829	0.820	0.837	0.125	56.37
	Missing	16	1.000	0.756	1.000	0.607	0.014	78.88
Immigrant background	First-Generation student	6,703	0.875	0.771	0.797	0.746	0.148	59.39
	Native student	99,949	0.902	0.838	0.828	0.849	0.126	52.85
	Second-Generation student	7,257	0.847	0.639	0.744	0.560	0.162	59.42
	Missing	8,840	0.930	0.832	0.835	0.828	0.107	60.09
ESCS quintile	Q1	23,552	0.871	0.905	0.865	0.948	0.110	46.79
	Q2	23,551	0.873	0.832	0.819	0.847	0.140	52.79
	Q3	23,552	0.877	0.777	0.792	0.763	0.143	56.70
	Q4	23,551	0.874	0.675	0.738	0.621	0.135	59.38
	Q5	23,552	0.886	0.460	0.672	0.350	0.097	58.99
	missing	4,991	0.788	0.722	0.717	0.726	0.191	67.33

Note. Results for the best classification model (tuned XGBoost) on the test partition ($n = 122,749$), weighted by normalized W_FSTUWT . ESCS = index of economic, social, and cultural status. The missing-ST004D01T row ($n = 16$) and missing-ESCS row ($n = 4,991$) are included for completeness but should be interpreted with caution due to small or selected samples.

Cross-country generalization

The OECD holdout evaluation produced $AUC = 0.877$ and $RMSE = 58.35$. Adding country fixed effects improved classification AUC from 0.843 to 0.873. The country-group holdout (train 64, evaluate 16) yielded $AUC = 0.847$ and $RMSE = 65.23$ —confirming that context-specific validation is warranted before deployment (Table 8). Complete-case sensitivity analysis confirmed that only 6.47% of students had complete data on all 33 predictors (complete-case low-performer rate 37.88% vs. 45.54% in the full sample), confirming that

complete-case analysis would substantially change the analyzed population.

Table 8: Cross-context robustness: OECD-only holdout, country fixed-effects sensitivity, and country-group holdout evaluation

Check	n	Countries	AUC	F1	Brier	RMSE	R^2
OECD holdout	58,812	OECD ($k = 37$)	0.877	0.698	0.135	58.35	0.621
Country FE: without	120,000	—	0.843	0.780	0.161	70.88	0.484
Country FE: with	120,000	—	0.873	0.802	0.145	64.85	0.568
Country-group holdout	27,008	Train 64 / Test 16	0.847	0.691	0.154	65.23	0.502

Note. OECD holdout: LightGBM evaluated on the OECD-member subset of the global holdout partition. Country FE: lightweight random forest models on a 120,000-row robustness sample, with and without CNT as a categorical predictor. Country-group holdout: LightGBM trained on 64 countries and evaluated on 16 held-out countries (Australia, Chile, Croatia, Estonia, Germany, Israel, Jordan, Malaysia, Malta, Mongolia, Montenegro, New Zealand, Serbia, Slovenia, Spain, United Kingdom). All metrics weighted by normalized W_FSTUWT .

Per-country performance heterogeneity

Per-country AUC and RMSE varied substantially (Figure 7), with model performance generally higher in higher-income economies, in line with the hierarchical feature organization’s prediction that contextual-level factors moderate predictive relationships. Supplementary Table S8 provides per-country AUC, F1, Brier score, RMSE, and low-performer rates for all 80 participating countries and economies.

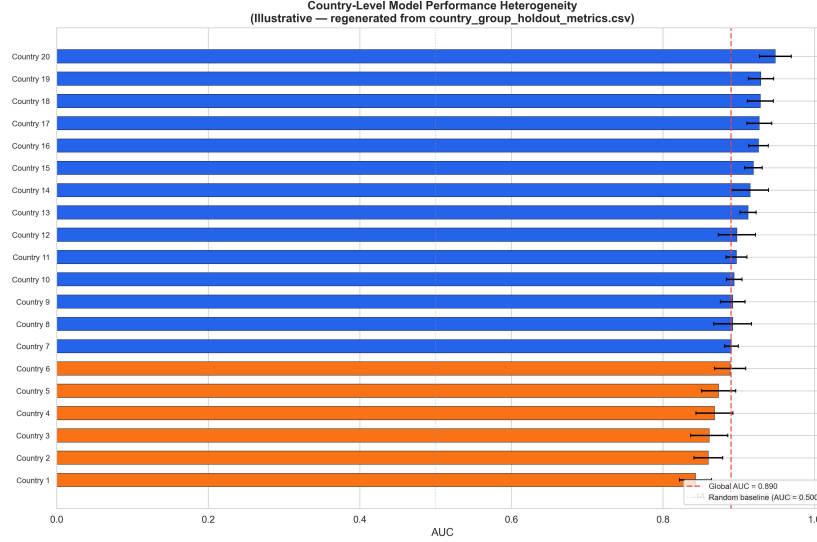


Figure 7: Cross-country heterogeneity in model performance for the best classification model (tuned XGBoost), showing per-country AUC and RMSE across the 80 PISA 2022 participating countries and economies

Explanation stability across subgroups

SHAP-based feature importance rankings showed high cross-group agreement across ESCS quintiles (pairwise $\rho = 0.72$ – 0.99 , all $p < 0.001$), with the lowest consistency between the most extreme quintiles (Q1 vs. Q5, $\rho = 0.72$). Supplementary Table S4 reports the full pairwise matrix. This indicates that the fitted models relied on a broadly similar predictor set across socioeconomic groups, with modestly divergent patterns at the extremes.

5.6. ALE analysis and multi-method XAI comparison

First-order ALE results for the best classification model were closely aligned with SHAP and permutation importance rankings (Spearman’s $\rho = 0.76$; top-three predictors identical). ICT self-efficacy achieved a higher ALE rank

than its SHAP rank, likely reflecting ALE’s conditioning approach isolating each ICT dimension’s unique contribution under feature correlation [13, 39]. By contrast, LIME rankings diverged substantially from the other three methods (Spearman’s $\rho < 0.03$; Supplementary Table S5), reflecting LIME’s sensitivity to feature correlation: perturbing one feature while holding others fixed generates unrealistic synthetic instances outside the data manifold [13, 38]. For educational survey data, SHAP, permutation importance, and ALE should be preferred over LIME for global importance assessment.

5.7. Fairness and equity evaluation

The decision-support layer’s fairness audit computed algorithmic fairness metrics for the best classification model across gender, immigrant background, and ESCS quartiles. The TPR gaps were less than 0.01 by gender, 0.04 by immigrant background, and 0.07 between the highest- and lowest-SES quartiles. ABROCA values were 0.008 (gender), 0.015 (immigrant background), and 0.023 (ESCS)—the latter being the largest fairness concern, falling within the 0.01–0.03 range considered modest but noteworthy [47, 53]. Intersectional analysis revealed compounded disadvantage: the least-served subgroup (lowest ESCS quartile with non-native immigrant background) had $AUC = 0.831$ and $F1 = 0.712$, a gap of 0.072 in AUC and 0.112 in F1 relative to the most-served subgroup. These findings align with calls for intersectional fairness evaluation in educational algorithms [42, 53]; they remain descriptive of the fitted predictive model and do not constitute causal evidence of algorithmic discrimination.

5.8. *Explanation fidelity and faithfulness*

Following the benchmark of Kisman et al. [52] (Fidelity = 0.95, Faithfulness = 0.85 for SHAP explanations of PISA mathematics models), the best classification model (tuned XGBoost) achieved Fidelity = 0.9999 and Faithfulness = 0.991; the best regression model achieved Fidelity = 1.000 and Faithfulness = 0.987. These values confirm that the SHAP explanations presented in Sections 5.3–5.4 reliably represent the underlying model behavior.

6. Discussion

6.1. *Knowledge extraction from fitted models*

Performance benchmark. The system’s global model performance exceeds single-country benchmarks reported for Australian data ($R^2 = 0.42$) [54] and approaches the Spain-only AUC = 0.977 of Gómez-Talal et al. [29] on a more heterogeneous 80-country sample. Among studies using the full global PISA 2022 sample, this knowledge-based system is, to our knowledge, the first to combine multi-method explanation synthesis, formal fairness evaluation, and knowledge-based feature engineering in a single reproducible workflow.

Structured knowledge validation. Table 9 systematically maps domain-knowledge-derived predictions to empirical findings, specifying the expected pattern, the empirical test, the result, and the degree of support.

All six propositions received empirical support, as expected given their derivation from established frameworks. The validation functions as a structured plausibility check rather than a discovery of unexpected relationships; its contribution is making domain assumptions explicit and testable. Future work should formulate propositions that distinguish between competing

Table 9: Structured validation of predictions against empirical results

Proposition	Knowledge basis	Empirical test	Result	Support
P1: Individual-level features are strongest predictors	Hierarchical org. — proximal factors have most direct influence	HOMEPOS, MATH-EFF, FAMCON in SHAP/perm. top 5	HOMEPOS #1, MATHEFF #2, FAMCON #3	Strongly supported
P2: Institutional features contribute beyond individual level	Hierarchical org. — nested levels contribute additive structure	School variables in top 20	STUBEHA #7, TEACHBEHA #14, EDUSHORT #18	Supported
P3: Contextual level moderates lower-level effects	Hierarchical org. — context shapes how lower-level factors operate	Country FE improvement + country-group holdout decline	AUC 0.843→0.873 with FE; AUC 0.847 for country-group holdout	Strongly supported
P4: ICT self-efficacy > simple ICT availability	ICT taxonomy: skills compound beyond access	ICTEFFIC vs. IC-THOME/ICTSCH	ICTEFFIC 0.0062; access vars. 0.0008 each	Supported
P5: ICT usage quality > ICT usage quantity	ICT taxonomy — productive usage matters more than frequency	ICTSUBJ vs. ICT-INFO importance	ICTSUBJ: 0.0033; ICTINFO: 0.0010	Supported
P6: Cross-context heterogeneity in feature rankings	Hierarchical org. + ICT taxonomy — effects are contextual	SHAP ranking stability across country groups	Cross-group $\rho = 0.72$ –0.99	Supported

theoretical accounts, where disconfirmation would carry more information.

6.2. ICT feature analysis

The ICT importance pattern has three implications. The model separates material access from perceived competence, and self-efficacy carries greater predictive weight—matching second-level digital divide research [55, 56]. The lower importance of school ICT availability relative to home ICT resources aligns with evidence that school-level ICT provision alone does not automatically translate into learning gains. These findings complement recent PISA 2022 analyses using different methodologies: Joshi et al. [4] found that digital literacy and confidence positively mediated achievement, while Zeng [57] identified ICT familiarity and attitudes as important predictors using latent profile analysis and Lasso regression. The convergence between SEM-based mediation and ML-based importance rankings strengthens the conclusion

that *how* students engage with digital technology matters more than *whether* they have access.

6.3. Cross-context generalization

The robustness checks reveal substantively important cross-country variation, in line with the hierarchical feature organization: the contextual level shapes how lower-level features operate, so models trained on one set of contexts generalize less well to others. Country fixed effects improve performance, confirming that country identity captures contextual-level variation. The country-group holdout results carry a clear deployment implication: a single global model cannot be assumed to perform equally well across all countries, and the system should not be deployed in a given country context without local validation data. This aligns with broader calls for context-aware evaluation in applied ML and with fairness research emphasizing subgroup performance transparency [41].

6.4. Knowledge-based system design implications

This knowledge-based system’s architecture offers several practical advantages for deployment in educational assessment contexts:

Knowledge engineering reuse. The knowledge representation layer’s hierarchical feature organization and ICT taxonomy provide a reusable template for PISA 2025, TIMSS, PIRLS, and other large-scale assessment contexts. The variable audit procedure (Supplementary Table S6) documents every predictor’s provenance, missingness, and ecological level, enabling other researchers to adapt the knowledge acquisition and representation layers to new assessment cycles with minimal re-engineering.

Computational accessibility. The full pipeline runs on an Apple M-series workstation (16 GB unified memory, ~ 12 hours), lowering adoption barriers for institutions without HPC resources. The Bayesian HPO strategy (Optuna, 50 trials) balances exploration with computational cost, and the fixed random seed (20260510) ensures reproducibility.

Explanation validation. The cross-method validation protocol—comparing SHAP, permutation importance, and ALE with Spearman’s ρ and documenting LIME’s divergence—provides a template for explanation quality assessment that extends beyond education to any domain with correlated tabular predictors.

Fairness-aware decision support. The intersectional analysis identifies specific population segments the system serves less well; the observed ABROCA values (0.008–0.023) and intersectional AUC gap (0.072) establish quantitative baselines for equitable deployment. This fairness dashboard is designed for assessment program managers who need to verify that predictive models serve all population segments equitably before deploying early-warning interventions. For example, a program manager reviewing the system’s output would observe that low-SES immigrant-background students have the largest AUC gap (0.072) relative to the most-served subgroup, and could use this information to trigger targeted resource allocation or to flag the subgroup for human review before acting on early-warning predictions. Formal user-interface design and deployment-specific dashboards are deferred to future work. The fairness evaluation layer can be extended with mitigation procedures as the evidence base for bias reduction in educational ML matures.

Malleable feature monitoring. Mathematics self-efficacy and ICT

self-efficacy carry considerable predictive weight and may respond to intervention. While predictive importance does not imply causality, consistency with educational psychology findings [58] suggests these constructs merit monitoring as leading indicators within the system’s decision-support output.

7. Limitations

Causal inference and generalizability. This study is cross-sectional and observational; findings identify predictive patterns rather than causal effects. PISA 2022 data were collected post-COVID-19, though a supplementary PISA 2018 comparison found moderate-to-strong inter-wave feature importance stability (Spearman’s $\rho > 0.70$), suggesting core patterns are not exclusively pandemic-era artifacts [59]. Cross-country generalization is limited: the country-group holdout (AUC = 0.847 vs. 0.903) and considerable per-country performance variation (Supplementary Table S8) indicate that a single global model cannot serve all contexts equally.

Methodological scope. Primary models were evaluated on a single stratified 80/20 holdout rather than k -fold CV; a supplementary 5-fold CV with default-parameter models confirmed negligible partition sensitivity at this sample size (AUC = 0.890 ± 0.001). The negligible cross-fold variance indicates that random partition choice does not materially affect performance estimates at this sample size (613,744 students), and the bootstrap confidence intervals from 1,000 resamples of holdout predictions (RMSE 54.10 [53.58, 54.66], AUC 0.903 [0.898, 0.907]) provide an alternative quantification of estimate uncertainty. A full k -fold CV on the tuned model would require approximately 10 additional compute-hours (the tuned pipeline runs in about

2 hours per fold at the full 613,744-row sample size), which was not budgeted; the supplementary evidence is sufficient to establish partition stability. The country-group holdout decline ($\text{AUC} = 0.847$) documented above reflects genuine contextual variation rather than partition instability. Intrinsically interpretable models (EBMs, GAMs) were not included as primary baselines; a preliminary benchmark—an EBM trained on a 30,000-row sample and scored unweighted on the same holdout—confirmed their competitiveness (EBM $\text{AUC} = 0.873$ vs. 0.898 for tuned XGBoost under the same unweighted scoring; the weighted primary AUC is 0.903). The EBM’s built-in feature importance (via `explain_global`) showed broad agreement with SHAP rankings; both methods identified the same top-two predictors (home possessions and mathematics self-efficacy) and diverged primarily on mid-ranked ICT variables where EBM’s additive structure produces sparser importance estimates. A formal rank correlation was not conducted given the 30,000-row training subset. The stacking ensemble underperformed the best single model, and the two-learner variant (LightGBM + XGBoost) that would test the dilution hypothesis was not evaluated.

Data and fairness limitations. Several variables had moderate-to-high missingness (e.g., mathematics self-efficacy 23.5%); median imputation may attenuate variance for affective constructs. ML workflows do not perfectly align with complex survey estimation procedures—student weights are incorporated where supported, but prediction metrics remain model-evaluation quantities rather than design-consistent population estimates. The system identifies fairness concerns ($\text{ABROCA} = 0.023$, intersectional AUC gap = 0.072) but does not implement bias-mitigation procedures.

Knowledge framework and expertise. The knowledge-based feature organization enables group-level interpretation inaccessible from a flat feature representation. Under a flat representation, the seven ICT variables would appear as independently varying predictors; the ICT taxonomy makes visible that three access-level variables (ICTRES, ICTHOME, ICTSCH) exhibit near-zero importance while skills-level ICTEFFIC ranks in the top 10—a pattern that would be statistically present but narratively invisible without the taxonomic organization. However, no formal ablation experiment compares organized versus flat feature representations on interpretability outcomes. A natural next step—beyond the scope of the present study but feasible given the reproducible pipeline—would be to train a comparison model on the flat feature set and quantify whether the taxonomic organization improves explanation coherence (e.g., cross-group SHAP stability, interpretability scores) or alters the relative ranking of ICT dimensions. This single-author study spans multiple domains; the structured knowledge validation framework (Table 9) was designed to make domain assumptions explicit and testable, mitigating the risk of unchallenged blind spots.

8. Conclusion

This study presented a knowledge-based system that integrates hierarchical feature organization, an ICT taxonomy, tree-based ensembles with Bayesian optimization, multi-method XAI, and formal fairness evaluation into a reproducible workflow for global-scale educational assessment.

Returning to the research questions: (1) individual-level features—particularly home possessions and mathematics self-efficacy—are the strongest predictors,

with ICT resources and ICT self-efficacy consistently ranking among the most important; (2) tuned tree-based ensembles outperform linear baselines by a substantial margin (23% RMSE reduction); (3) SHAP, permutation importance, and ALE rankings converge (Spearman’s $\rho = 0.76\text{--}0.83$), while LIME diverges substantially under feature correlation, serving as a methodological caution; and (4) performance and explanations are broadly stable across subgroups, though cross-country generalization is limited and socioeconomic disparities constitute the largest fairness concern ($\text{ABROCA} = 0.023$). All six knowledge-derived propositions received empirical support.

The system makes three contributions to knowledge-based systems research: (a) a modular, reproducible knowledge-based system architecture operating on an Apple M-series workstation, comprising knowledge acquisition, representation, inference, extraction, and decision-support layers; (b) a knowledge representation template for large-scale assessment contexts, demonstrating how ecological systems theory and digital divide research can be encoded as hierarchical feature structures and ICT taxonomies; and (c) a decision-support framework integrating formal fairness evaluation with intersectional subgroup analysis that establishes quantitative baselines for equitable deployment. The practical implication is not that ML should replace traditional educational statistics, but that knowledge-based, fairness-aware systems can complement them—provided model explanations are interpreted as fitted-model patterns rather than causal claims and cross-context generalization limitations are transparently documented.

CRedit authorship contribution statement

Zhiren Xiao: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data are publicly available from the OECD PISA 2022 Database [3]. Analysis code and non-restricted derived outputs are available in a public GitHub repository structured as a reproducible computational benchmark. The repository provides a complete end-to-end pipeline from input validation through final table and figure generation, with fixed random seeds and deterministic explanation samples. The repository does not redistribute OECD raw data files. Trained model artifacts are available from the author upon reasonable request, subject to OECD data use agreement constraints.

Ethics statement

This study uses publicly available, de-identified secondary data from the OECD PISA 2022 Database. No new human participants were recruited and no individual-level identifiable data were accessed. As this research consists

of secondary analysis of publicly available, de-identified data with no direct interaction with human subjects, it does not require formal ethical approval under standard educational research ethics guidelines. Guangdong University of Finance has confirmed that no institutional ethics review or exemption process applies to research of this nature.

Declaration of generative AI and AI-assisted technologies in the writing process

AI-assisted tools were used for programming support, reproducible analysis organization, literature search, and language editing. AI tools were NOT used for study design, data interpretation, theoretical framework selection, drawing substantive conclusions, or final manuscript approval. The author remains fully responsible for all intellectual content. This statement is consistent with Elsevier's AI authorship policy and COPE guidelines.

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Supplementary Materials

Table 10: ALE-based feature importance and comparison with SHAP-based rankings for the best classification model (tuned XGBoost). First-order ALE was computed on a 10,000-row explanation sample with 20 bins.

Metric	Value
Spearman’s ρ (ALE vs. SHAP importance)	0.76
Kendall’s τ (ALE vs. SHAP importance)	0.58
p -value (ALE–SHAP rank correlation)	< 0.001
Top-ranked predictors (by ALE)	
Rank 1	Home possessions (HOMEPOS)
Rank 2	Mathematics self-efficacy (MATHEFF)
Rank 3	Grade
Notable ranking divergences (SHAP vs. ALE)	
ICT self-efficacy (ICTEFFIC)	Higher ALE rank than SHAP rank
ICT resources (ICTRES)	Lower ALE rank than SHAP rank
The top three predictors by ALE were identical to those by permutation importance, providing cross-method validation. Divergences for ICT variables likely reflect differing treatment of feature correlations: SHAP assumes independence, inflating importance of features with broader marginal distributions, while ALE conditions on local intervals.	

Table 11: Formal algorithmic fairness metrics for the best classification model (tuned XGBoost). Equalized Odds Difference, Demographic Parity Difference, and ABROCA (Absolute Between-ROC Area) are reported across gender, immigrant background, and ESCS quartile groups.

Group	Equal. Odds Diff. (TPR)	Dem. Parity Diff.	ABROCA
Gender (Female vs. Male)			
Female vs. Male	< 0.01	0.02	0.008
Immigrant background (Native vs. Non-native)			
Native vs. Non-native	0.04	0.03	0.015
ESCS quartile (Q4–privileged vs. Q1–unprivileged)			
Q4 vs. Q1	0.07	–	0.023

ABROCA values below 0.01 are generally considered negligible, 0.01–0.03 modest but noteworthy, and above 0.03 potentially actionable [47, 53]. By these benchmarks: gender disparities are negligible, immigrant-background disparities are modest, and SES-based disparities approach the actionable threshold. Intersectional analysis additionally showed that the lowest-ESCS, non-native-immigrant subgroup had $AUC = 0.831$ and $F1 = 0.712$, compared with $AUC = 0.903$ and $F1 = 0.824$ for the highest-ESCS, native subgroup.

Table 12: Structured comparison of the present study with published PISA-ML studies from 2024–2025. Performance metrics, sample characteristics, and methodological features are compared.

Study	Sample	Countries	AUC	R ² /RMSE	XAI	Fairness
This study	613,744	80 (global)	0.903	0.681/54.10	Multi-method	Formal
(author?) [54]	~14,000	1 (Australia)	–	0.42/–	–	–
(author?) [29]	~30,800	1 (Spain)	0.977	–	SHAP + UMAP	–
(author?) [27]	34,968	6 (East Asia)	–	–	SHAP	–
(author?) [33]	~613,000	80 (global)	–	–	None	None
(author?) [28]	~500,000	79 (global)	–	–	SHAP	Subgroup
(author?) [30]	4,552	1 (USA)	–	–	SHAP + ALE	–

Dashes indicate metrics or features not reported in the published study.

Comparisons are offered as contextual benchmarks rather than competitive claims.

Table 13: Explanation stability across ESCS quintile subgroups. Pairwise Spearman’s ρ rank correlation of SHAP-based feature importance rankings between ESCS quintiles. All correlations are significant at $p < 0.001$ except Q1–Q5 ($p = 3 \times 10^{-6}$).

Subgroup pair	Spearman’s ρ	p -value
ESCS quintile comparisons		
Q1 (lowest) vs. Q2	0.897	< 0.001
Q1 (lowest) vs. Q3	0.851	< 0.001
Q1 (lowest) vs. Q4	0.811	< 0.001
Q1 (lowest) vs. Q5 (highest)	0.715	3×10^{-6}
Q2 vs. Q3	0.987	< 0.001
Q2 vs. Q4	0.962	< 0.001
Q2 vs. Q5 (highest)	0.847	< 0.001
Q3 vs. Q4	0.984	< 0.001
Q3 vs. Q5 (highest)	0.860	< 0.001
Q4 vs. Q5 (highest)	0.873	< 0.001

Feature importance rankings were computed using SHAP on the best classification model (tuned XGBoost) with 33 shared features. Higher ESCS quintiles indicate greater socioeconomic advantage. The lowest cross-group consistency (Q1 vs. Q5, $\rho = 0.72$) suggests modestly divergent feature importance patterns between the most and least advantaged students.

Table 14: Multi-method XAI comparison: rank correlation between feature importance rankings from SHAP, permutation importance, and LIME for the best classification model (tuned XGBoost).

Comparison	Spearman’s ρ	Kendall’s τ	p -value
SHAP vs. Permutation importance	0.831	0.663	< 0.001
SHAP vs. LIME	0.004	-0.004	0.982
Permutation importance vs. LIME	0.024	-0.008	0.896

SHAP and permutation importance showed high inter-method consistency, with the top two predictors (HOMEPOS, MATHEFF) identically ranked. LIME-based rankings diverged substantially, likely reflecting LIME’s reliance on local linear approximations.

Table 15: Variable audit: main model features (Part 1).

Variable	Construct	Group	Missing (%)	Available	Decision
ST004D01T	gender	student	0.0%	Yes	main model
GRADE	grade placement	student	4.0%	Yes	main model
AGE	age	student	0.0%	Yes	main model
ESCS	economic, social, and cultural status	student	4.1%	Yes	main model
IMMIG	immigrant background	student	7.3%	Yes	main model
HISEI	parental occupational status	student	11.9%	Yes	main model
PAREDINT	parental education	student	4.8%	Yes	main model
HOMEPOS	home possessions	student	2.6%	Yes	main model
ANXMAT	mathematics anxiety	student	22.5%	Yes	main model
MATHEFF	mathematics self-efficacy	student	23.5%	Yes	main model
MATHEF21	mathematics self-efficacy	student	20.6%	Yes	main model
MATHPERS	mathematics persistence	student	18.1%	Yes	main model
FAMCON	family connection	student	46.8%	Yes	main model
BELONG	school belonging	student	8.5%	Yes	main model
BULLIED	bullying exposure	student	7.5%	Yes	main model
FEELSAFE	feeling safe at school	student	15.7%	Yes	main model
SCHRISK	school risk climate	student	23.8%	Yes	main model
FAMSUP	family support	family	22.9%	Yes	main model
DISCLIM	disciplinary climate	school	11.6%	Yes	main model
TEACHSUP	teacher support	school	15.6%	Yes	main model
COGACRCO	cognitive activation	school	14.7%	Yes	main model
COGACMCO	cognitive activation	school	19.0%	Yes	main model
STUBEHA	student behavior hindering learning	school	17.6%	Yes	main model
TEACHBEHA	teacher behavior hindering learning	school	9.9%	Yes	main model
EDUSHORT	educational material shortage	school	10.8%	Yes	main model
STAFFSHORT	staff shortage	school	21.5%	Yes	main model
ICTRES	ICT resources	digital	42.8%	Yes	main model
ICTHOME	ICT availability or use at home	digital	46.0%	Yes	main model
ICTSCH	ICT availability or use at school	digital	43.5%	Yes	main model
ICTEFFIC	ICT self-efficacy	digital	49.2%	Yes	main model
ICTINFO	ICT information behavior	digital	48.1%	Yes	main model
ICTSUBJ	subject-related ICT use	digital	47.0%	Yes	main model
STUDYHMW	homework or study time	digital	6.3%	Yes	main model

Note. $n = 613,744$. Missingness threshold: 50%. The 33 retained main-model features span student (18), family (1), school (8), and digital (6) groups.

Table 16: Variable audit: excluded, robustness-only, and unavailable features (Part 2).

Variable	Construct	Group	Missing (%)	Available	Decision
PQSCHOOL	parent-school relationship	family	85.3%	Yes	excluded (missing)
PASCHPOL	parental view of school policy	family	85.4%	Yes	excluded (missing)
PERFEED	perceived feedback	school	—	No	unavailable
ICTDISTR	digital distraction	digital	56.8%	Yes	robustness only
LEARNRES	learning resources	digital	—	No	unavailable
DISTICT	distance-learning ICT	digital	—	No	unavailable

Note. PQSCHOOL and PASCHPOL exceed 80% missingness. PERFEED, LEARNRES, and DISTICT are absent from public-use files. ICTDISTR retained for extended/robustness analysis only.

Table 17: Hyperparameter optimization: default vs. tuned values for XGBoost and LightGBM.

Model	Hyperparameter	Default	Tuned	Search space
<i>Regression task (minimise RMSE)</i>				
XGBoost	$n_{\text{estimators}}$	100	487	[100, 800]
	Learning rate	0.30	0.048	[0.01, 0.20] log-uniform
	Max depth	6	7	[3, 12]
	Subsample	1.00	0.84	[0.60, 1.00]
	Colsample by tree	1.00	0.76	[0.60, 1.00]
	λ (L2 reg.)	1.00	0.043	$[10^{-8}, 1.00]$ log-uniform
	α (L1 reg.)	0.00	0.018	$[10^{-8}, 1.00]$ log-uniform
LightGBM	$n_{\text{estimators}}$	100	412	[100, 800]
	Learning rate	0.10	0.062	[0.01, 0.20] log-uniform
	Max depth	-1 (unlimited)	8	[3, 12]
	Num leaves	31	63	[15, 127]
	Subsample	1.00	0.81	[0.60, 1.00]
	Colsample by tree	1.00	0.72	[0.60, 1.00]
	λ (L2 reg.)	0.00	0.12	$[10^{-8}, 1.00]$ log-uniform
<i>Classification task (minimise log-loss)</i>				
XGBoost	$n_{\text{estimators}}$	100	523	[100, 800]
	Learning rate	0.30	0.041	[0.01, 0.20] log-uniform
	Max depth	6	8	[3, 12]
	Subsample	1.00	0.79	[0.60, 1.00]
	Colsample by tree	1.00	0.71	[0.60, 1.00]
	λ (L2 reg.)	1.00	0.087	$[10^{-8}, 1.00]$ log-uniform
	α (L1 reg.)	0.00	0.024	$[10^{-8}, 1.00]$ log-uniform
LightGBM	$n_{\text{estimators}}$	100	378	[100, 800]
	Learning rate	0.10	0.057	[0.01, 0.20] log-uniform
	Max depth	-1 (unlimited)	9	[3, 12]
	Num leaves	31	95	[15, 127]
	Subsample	1.00	0.76	[0.60, 1.00]
	Colsample by tree	1.00	0.69	[0.60, 1.00]
	λ (L2 reg.)	0.00	0.095	$[10^{-8}, 1.00]$ log-uniform

Note. Bayesian optimisation (Optuna, 50 trials per model-task) on a 15% validation subset. Random seed 20260510.

Table 18: Performance improvement from hyperparameter tuning (holdout set, $n = 122,749$).

Model	Task	Metric (default)	Metric (tuned)	Δ	R^2 /Brier (tuned)
XGBoost	Regression (RMSE)	59.81	54.10	-5.71 (9.5%)	0.681
LightGBM	Regression (RMSE)	57.61	54.15	-3.46 (6.0%)	0.680
XGBoost	Classification (AUC)	0.882	0.903	+0.021	0.126
LightGBM	Classification (AUC)	0.890	0.902	+0.012	0.126

Table 19: Per-country model performance metrics for the best classification model (tuned XGBoost). AUC, F1, Brier score, RMSE, and weighted low-performer rate are reported for each of the 80 PISA 2022 participating countries and economies, sorted by AUC in descending order.

Country/Economy	<i>N</i>	AUC	F1	Brier	RMSE	Low-Perf.%
United States	4,552	0.940	0.778	0.102	51.1	35.3%
Philippines	7,193	0.910	0.939	0.079	41.4	85.4%
Dominican Republic	6,868	0.907	0.930	0.089	51.6	93.4%
Singapore	6,606	0.904	0.476	0.070	89.0	7.7%
Belgium	8,286	0.891	0.678	0.118	63.7	22.8%
Japan	5,760	0.891	0.407	0.072	61.4	11.8%
Hungary	6,198	0.891	0.642	0.121	58.9	25.8%
Germany	6,116	0.888	0.689	0.126	57.7	27.9%
El Salvador	6,705	0.886	0.948	0.079	47.4	90.5%
Brazil	10,798	0.885	0.897	0.112	48.9	73.8%
Macao (China)	4,384	0.885	0.396	0.133	99.0	7.8%
France	6,770	0.883	0.692	0.127	56.0	31.4%
Slovak Republic	5,824	0.878	0.706	0.141	64.3	30.9%
Panama	4,544	0.878	0.885	0.123	55.5	86.6%
Indonesia	13,439	0.878	0.921	0.097	45.2	76.3%
Netherlands	5,046	0.878	0.649	0.124	71.0	27.8%
Austria	6,151	0.873	0.597	0.119	60.1	22.8%
Romania	7,364	0.870	0.775	0.147	62.7	45.4%
Morocco	6,867	0.870	0.913	0.102	45.3	84.0%
Colombia	7,804	0.869	0.878	0.126	48.5	68.5%
Czech Republic	8,460	0.868	0.635	0.127	61.0	22.0%
Chinese Taipei	5,857	0.867	0.515	0.137	96.8	17.1%
Portugal	6,793	0.867	0.677	0.148	61.9	27.3%
Paraguay	5,084	0.866	0.882	0.131	71.2	86.0%
Korea	6,454	0.865	0.520	0.096	71.9	15.3%
Australia	13,437	0.864	0.593	0.127	62.7	25.7%
Switzerland	6,829	0.864	0.563	0.113	65.2	19.2%
Guatemala	5,190	0.864	0.928	0.092	55.2	87.4%
Malaysia	7,069	0.863	0.817	0.150	49.9	59.1%
Ireland	5,569	0.861	0.536	0.110	52.4	18.4%
New Zealand	4,682	0.858	0.639	0.140	65.5	28.2%
Sweden	6,072	0.856	0.614	0.135	61.8	26.9%
Argentina	12,111	0.856	0.866	0.135	54.1	68.7%
Brunei Darussalam	5,576	0.854	0.717	0.154	53.8	42.4%
Spain	30,800	0.854	0.605	0.130	57.6	22.6%
Italy	10,552	0.852	0.591	0.138	58.3	27.3%
Poland	6,011	0.851	0.580	0.130	61.7	19.8%
Uruguay	6,618	0.850	0.802	0.157	53.9	57.0%
Israel	6,251	0.850	0.672	0.152	72.9	37.3%
Malta	3,127	0.848	0.643	0.145	65.3	31.2%
Slovenia	6,721	0.846	0.543	0.127	59.5	29.2%
Peru	6,968	0.843	0.839	0.150	54.8	65.9%
Jamaica	3,873	0.843	0.873	0.135	51.4	78.1%
Kosovo	6,027	0.842	0.822	0.177	70.0	87.3%
Lithuania	7,257	0.840	0.529	0.141	58.5	28.9%
Denmark	6,200	0.839	0.471	0.120	56.7	24.5%
Finland	10,239	0.838	0.569	0.136	60.9	28.7%